

Maintenance Schedule

Frac Pack

MTU Series 4000 12 V 4000 T94L

PFRPK12V4000-1A0

Oil & Gas

Application Group 4D

MS50093/02E



Power. Passion. Partnership.

Table 1: applicable for

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

1 Maintenance Schedule

1.1 Preface

MTU maintenance concept

The following applies in the case of engines which are subject to emissions regulations:

Emissions regulations prohibit alteration, removal or addition of any mechanical or electronic component or calibration that could affect the emissions characteristics of the engine. Maintenance, replacement, or repair of emissions control devices, sensors, cylinder heads and exhaust flaps, for example, can only be performed using MTU approved components or equivalents.

Failure to adhere to these guidelines could result in violation of legal requirements.

MTU Maintenance Schedules ensure the reliability and performance of MTU engines and must be adhered to during the full life of the engine. In the case of engines which are not subject to any emissions regulations, the maintenance intervals cited are to be regarded as guideline values.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. The limit which occurs first is applicable. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.

QL2: Component exchange (corrective only).

QL3: Maintenance work which requires partial disassembly of the engine.

QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:
The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).
- Battery:
Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not included in the MTU supply scope; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application Group

4D	Frac-operation (continuous operation with low load)
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Load factor

Load factor	<35%				
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Load profile

Load	90%	35%	20%			
Corresponding operating time	7%	77%	16%			

1.2 Maintenance schedule matrix

0-10,500 Operating hours

Item	Limit	Operating hours [h]																						
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	
Engine																								
Emergency air shut-off flaps	1 a																							
Engine oil filter	2 a																							
Centrifugal oil filter	2 a																							
Engine operation	-	X																						
Fuel filter	2 a			X		X		X		X		X		X		X		X		X		X		
Belt drive	2 a			X		X		X		X		X		X		X		X		X		X		
Primary fuel filter	2 a			X		X		X		X		X		X		X		X		X		X		
Valve gear	2 a							X						X						X				
Coolant filters	2 a							X						X						X				
Exhaust gas recirculation	2 a							X						X						X				
NOx sensor	6 a										X									X				
Lambda sensor	6 a										X									X				
Fuel injectors / injectors	9 a										X									X				
Engine governor	9 a										X									X				
Combustion chambers	6 a										X									X				
Engine mounts	9 a										X									X				
Turbochargers	18 a										X									X				
Exhaust gas recirculation	6 a										X									X				
Engine mounts	5 a										X									X				
Turbochargers	18 a																			X				
Component maintenance	18 a																			X				
Crankcase breathers	6 a																			X				
Fuel delivery pump	18 a																			X				
Battery charging generator	18 a																			X				
Cylinder heads	18 a																			X				
Hose lines	6 a																			X				
Extended component maintenance	18 a																			X				
Engine mounts	18 a																			X				
Transmission																								
Transmission operation	-	X																						
Gear oil filter	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Transmission mounts	-			X		X		X		X		X		X		X		X		X		X		
Screw connections	-			X		X		X		X		X		X		X		X		X		X		
Connectors/wiring harness	-			X		X		X		X		X		X		X		X		X		X		
Torsional resilient coupling	6 m					X				X				X				X				X		
Suction strainer (magnetic)	-																			X				
Transmission	-																			X				
Preheater / Jacket water heater																								
Preheater / Jacket water heater	1 a																							
Fuel pump	5 a																							
Circulation pump	-																							
w = weeks m = months a = years																								